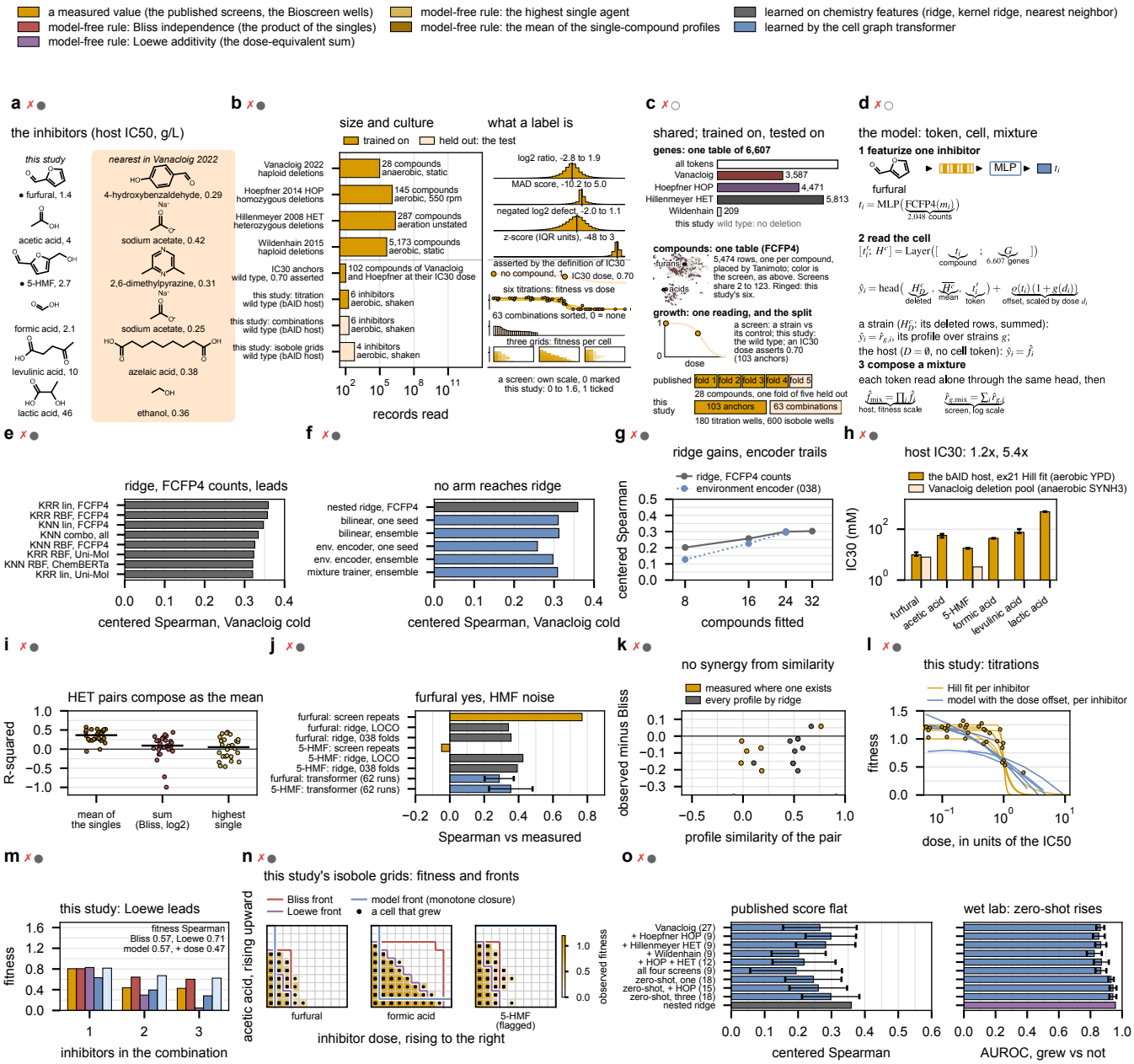


Figure 2: Composition is what the wet lab buys, and no published screen lifts the published score. The six hydrolysate inhibitors are met by one wild-type host; Vanacloig 2022 profiles two of them itself, and the nearest profiled compound of each of the other four sits beside it (**a**). Four published deletion screens of three genotypes and this study's wild-type wells differ in size, culture and in what a label is, a strain-level score on a screen's own scale against a growth rate per well (**b**); what joins them is one gene table, one compound table and one growth reading, with a screen's IC30 dose asserting the wild type at 0.700 (**c**). The cell graph transformer reads one token per inhibitor through one head, the host as the empty genotype at the gene mean, the dose as a gain on the medium's offset, and composes the readings as Bliss independence (**d**). On the published side nested ridge on fingerprint counts, 0.359, is not beaten by any transformer arm (**e**, **f**), ridge gains about 0.049 per doubling of fitted compounds with the environment encoder below it at every count (**g**), and adding screens leaves the compound-cold score flat (**o**, left). On the wet-lab side Loewe additivity from the titrations orders the 63 combinations at a growth AUROC of 0.958 and Bliss from the measured singles at 0.872; the model trained on the anchors and the titration reads 0.862 on combinations it never saw and its front sits inside the isobole grids (**m**, **n**), and the zero-shot arms, which never see a titration well, rise with every screen added, inside the run spread (**o**, right). Three readings qualify the join. The host needs 1.23 and 5.42 times the screen's IC30 of furfural and 5-HMF (**h**); a screen's label is a strain's growth against its own control on its own scale, this study's a wild-type growth rate with no growth in a share of 0.650 of the combination wells (**b**); and the published profiles do not reach the private question: pairs compose as the mean of their singles (**i**), furfural's profile is predicted at 0.341 where its screen repeats at 0.769 while 5-HMF's screen does not repeat (**j**), and the similarity of two inhibitors' profiles does not anticipate their pair's departure from Bliss, Spearman 0.204 at p 0.462 (**k**). The titrations themselves (**l**) are read by the model only with the dose offset, Spearman 0.890 against 0.452 without it. Color is the source of a value (key above the sheet); dark and light fills in **b** and **c** are trained on and held out. Public scores are compound-cold centered Spearman, median over held-out compounds; wet-lab scores are the served growth call.